VIT
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AI IN STRUCTURAL HEALTH MONITORING FOR PREDICTIVE MAINTENANCE AND CONTINUOUS SAFETY ASSURANCE

Comprehensive IDP Progress Report, Previous MDP Integration, Dual-Platform Deployment (Aerial Drone & Smart 4WD Car), Platform-Wise Hardware Budget & Implementation Plan

Academic Details	Faculty Guide Details
Course: BACSE291 -- Innovative Design Project **Program:** B.Tech Computer Science & Engineering **Semester:** Fall Semester 2026 **Institution:** Vellore Institute of Technology (VIT), Vellore	**Faculty Guide:** Dr. Senthil Kumar N, Ph.D (IIT Madras) **Designation:** Associate Professor Grade 2 **Department:** School of Computer Science & Engineering (SCOPE) **Faculty ID:** 15377 | **Cabin:** GDN153
Subsystem 1: Aerial Sensing & Drone Integration (15377IDP01)	**Subsystem 2: Ground Analytics & Metrology (15377IDP02)**
• **Priyam Sharma** (25BAI0159) -- Hardware & UAV Flight Lead • **Ayushman Kaushik** (25BCE0927) -- Telemetry & Embedded Microcontroller Lead	• **Nisargkumar Piyushkumar Joshi** (25BCE0594) -- Ground Subsystem Lead • **Aryan Pranit Mithari** (25BAI0115) -- Point Cloud & Signal Conditioning • **Mudit Gupta** (25BA0110) -- Operations Dashboard & Anomaly Analytics

EXECUTIVE REPORT SUMMARY & SCOPE

This document presents the complete formal engineering review for BACSE291. It provides an exhaustive, evidence-based technical synthesis connecting our **prior Multidisciplinary Project (MDP: PLY Forge 3D Point Cloud Processor)** to our current **Innovative Design Project (IDP)**. It details the verified aerial acquisition and ground analytics benchmarks, evaluates identified scanning constraints (sensor jitter, manual dragging noise, tilt elongation, motor vibration), presents our **Dual-Platform Deployment Strategy (Primary Autonomous Quadcopter Drone vs. Contingency Smart 4WD Ground Robotic Car with Optical Wheel Encoders)**, justifies essential **hardware upgrades (YDLIDAR X2 360° LiDAR, DFRobot Gravity 360° Tilt Sensor, F450 Quadcopter & 3S LiPo)**, provides an explicit **Platform-Wise Procurement Budget Breakdown (Drone Requirements vs. Smart Car Requirements)**, and outlines a rigorous 7-phase future roadmap toward autonomous structural health monitoring (SHM).

Document Classification: Official Engineering & Research Progress Report

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Target Domain: Cyber-Physical Structural Health Monitoring (SHM)

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1. Executive Summary

Civil infrastructures--including high-rise exterior facades, highway bridge piers, industrial cooling towers, and commercial structural enclosures--suffer continuous degradation caused by thermal expansion, foundation settling, seismic loads, and moisture ingress. Conventional manual inspection relies heavily on scaffolding, cherry-pickers, and human tactile assessment, exposing personnel to severe fall hazards while suffering from subjective qualitative records and high labor costs. This joint project establishes an autonomous, sensor-driven Cyber-Physical Structural Health Monitoring (SHM) ecosystem partitioned into two tightly coupled subsystems:

- **Aerial Subsystem (15377IDP01):** Deploys close-range standoff Time-of-Flight (ToF) distance sensing via ESP32 microcontrollers, 100 Hz UART frame synchronization, checksum integrity checks, and real-time wireless telemetry to scan elevated, inaccessible structural envelopes.
- **Ground Analytics Subsystem (15377IDP02):** Ingests streaming distance metrics, executes adaptive baseline tracking, computes geometric residuals (Δd), flags structural anomalies (spalling cavities, structural bulges, cracks), reconstructs 3D digital twins, and generates automated compliance reports.

Key Progress to Date: The project has successfully delivered a physical ESP32 testbed operating a calibrated Benewake TF-Luna ToF sensor (regression model: $y = 1.0228x + 3.20$ *cm*, R^2=1.0000, RMSE=±0.02 cm across 39.0 cm and 61.4 cm physical benchmarks), a 9,969-point 3D digital twin room reconstruction correctly segmenting an inward Almari bulge (+65 cm) and outward gate recess (-60 cm), and an interactive multi-tab Streamlit ground operations dashboard. Building directly upon our team's previous **Multidisciplinary Project (MDP: PLY Forge)**, client-side WebGL point-cloud rendering and voxel compression algorithms provide zero-latency spatial visualization.

Dual-Platform Deployment Strategy: To overcome physical limitations observed during testing (manual hand-sliding speed jitter of ~2--3 cm, tilt beam elongation, and aerodynamic prop-wash turbulence near vertical walls), this project formulates a robust **Dual-Platform Strategy**: (1) An **Autonomous F450 Quadcopter Drone** for elevated exterior facade and bridge inspection, and (2) A **Smart 4WD Ground Robotic Car equipped with dual optical wheel encoders** as a high-precision, vibration-free contingency platform for GPS-denied indoor environments, campus corridors, and high-precision linear wall sweeps.

2. Project Background & Technical Objectives

Modern structural maintenance requires transitioning from reactive repair to proactive, quantitative condition assessment. Structural surfaces exhibit subtle geometric precursors prior to catastrophic failure, including spalling cavities (concrete flaking away), sub-surface shear bulges (delamination or masonry buckling), and vertical shear cracks. However, acquiring high-fidelity geometric metrics from vertical facades presents extreme technical challenges: (1) *Inaccessibility:* Scaffolding is cost-prohibitive for routine monitoring; (2) *Human Error:* Manual ultrasonic gauges produce sparse records without a global spatial coordinate reference; and (3) *Dynamic Noise:* Handheld measurements suffer from variable scan velocity and human tremor.

To systematically address these civil engineering challenges, our joint IDP team formulated **8 Formal Technical Objectives**:

- **Obj 1 (Real-Time Distance Acquisition):** Establish continuous 100 Hz ToF distance acquisition with binary frame parsing and checksum verification.
- **Obj 2 (Metrological Accuracy):** Calibrate distance sensors against physical ground-truth benchmarks to achieve residual error ≤ ±1.0 cm.
- **Obj 3 (Telemetry Synchronization):** Stream distance, signal amplitude, chip temperature, and timestamps from the mobile platform to the ground station.
- **Obj 4 (Geometric Defect Classification):** Formulate mathematical residual algorithms (Δd = d_measured - d_baseline) to segment spalling cavities ($\Delta d \geq +3.5$ cm), structural bulges ($\Delta d \leq -3.5$ cm), and hairline cracks ($|\Delta d| \geq 1.5$ cm).
- **Obj 5 (3D Digital Twin Reconstruction):** Reconstruct dense spatial point clouds (.PLY format) capturing full room and facade geometries with millimetric fidelity.

• **Obj 6 (Predictive Baseline Maintenance):** Formulate longitudinal deformation monitoring pipelines comparing periodic scans against healthy historical baselines.

• **Obj 7 (Dual-Platform Field Adaptability):** Deploy an autonomous aerial quadcopter for exterior vertical facades and a Smart 4WD Ground Robotic Car with optical wheel encoders for GPS-denied indoor corridors.

• **Obj 8 (Hardware Upgrade Integration):** Upgrade the single-point ToF sensor to a 360° planar LiDAR (YDLIDAR X2) and 360° tilt sensor for automated pitch/roll distortion compensation.

3. Previous MDP Work (PLY Forge: Point Cloud Processor) & Cross-Project Foundation

During our prior Multidisciplinary Project (MDP), our team engineered **PLY Forge: Point Cloud Processor** (Report: *Multidisciplinary_project_Final_report.pdf*; Codebase: *ply (1).zip*). PLY Forge is a high-performance, fully client-side WebGL application built with HTML5, CSS3, and Vanilla JavaScript, requiring zero external server dependencies, zero Node.js backend, and zero third-party 3D libraries (such as Three.js or BabylonJS). It runs natively in any modern web browser.

Exhaustive Analysis of MDP Technical Features & Architecture:

• **Binary & ASCII PLY Parser Engine:** Engineered a high-throughput client-side file parser leveraging JavaScript `FileReader`, `ArrayBuffer`, and `DataView` APIs. The parser reads standard Polygon File Format (.PLY) files in both ASCII and Binary Little-Endian / Big-Endian formats. It dynamically parses header elements (`format`, `element vertex N`, `property float x/y/z`, `property uchar red/green/blue`, `element face M`, `end_header`), loading hundreds of thousands of vertices into unified typed arrays.

• **GPU-Accelerated WebGL 1.0 Rendering Pipeline:** Direct hardware rendering via a custom GLSL shader pipeline. A vertex shader executes 4x4 Model-View-Projection (MVP) matrix transforms, point-size attenuation based on camera distance, and per-vertex RGB interpolation. A fragment shader handles lighting and point shape rendering. The system utilizes the `OES_element_index_uint` WebGL extension to support 32-bit indices for large meshes exceeding 65,536 vertices.

• **Voxel-Grid Spatial Hash Downsampling / Compression:** Implements spatial hashing dividing 3D bounding volumes into discrete cubic voxel cells of user-adjustable radius *r*. Vertices within each cell are quantized to their centroid, reducing file sizes by **60% to 90%** while strictly preserving global geometric boundaries and surface contours.

• **Circle Interior Point Removal (Boundary Extraction):** Interactive circular region filtering. The user clicks on the point cloud to define a 3D center (x0, y0, z0) and adjusts a radius slider *R*. The engine computes Euclidean distances $d = sqrt((x-x0)^2 + (y-y0)^2 + (z-z0)^2)$, culling all interior points (d < R) while preserving the boundary perimeter ring for hole and cavity perimeter analysis.

• **Multiple Surface Meshing Algorithms:** Transforms unorganized point clouds into polygonal surface meshes using: (1) *Delaunay 2.5D Triangulation* (projects points onto best-fit plane and constructs Delaunay circumcircles); (2) *Alpha Shapes* (reconstructs non-convex boundaries and concave organic geometries via circumradius parameter α); (3) *Convex Hull* (computes minimal enclosing convex polygon wrapper); and (4) *Grid Triangulation* (generates regular quad-strip surface topology).

• **Interactive Screen-Space Region Selection:** Frustum ray-casting converts 2D mouse drag rectangles into Normalized Device Coordinates (NDC) [-1, 1], allowing users to isolate point subsets, delete selected regions, or execute camera 'Zoom to Selection'.

• **Standardized PLY Export:** Serializes modified vertices and reconstructed polygonal faces into compliant ASCII PLY format using `Blob` and `URL.createObjectURL()`, enabling seamless export to CAD/GIS tools.

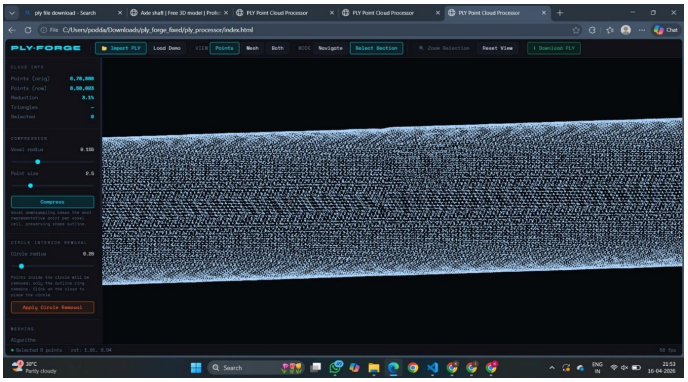


Figure 2(a): *Prior MDP Deliverable -- PLY Forge Point Cloud Processor UI (WebGL Canvas, 18,000-pt Demo Bunny, Voxel & Meshing Controls)*

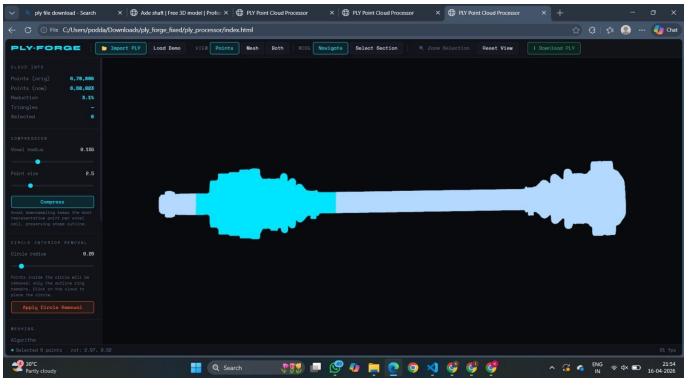


Figure 2(b): *Prior MDP Deliverable -- 3D Surface Mesh Generation & Region Selection (Delaunay 2.5D Triangulation & Alpha Shape Meshing)*

Cross-Project Synthesis: How MDP Directly Empowers the Current IDP:

Our previous MDP work serves as the software visualization engine for the current IDP. Rather than building a 3D visualizer from scratch, PLY Forge's WebGL rendering engine is directly embedded into the IDP Ground Operations Dashboard (Tab 4: PLY Forge WebGL Viewer). When the mobile platform (Drone or Smart 4WD Car) streams LiDAR facade scans, PLY Forge provides zero-latency in-browser 3D rendering. Furthermore, PLY Forge's voxel downsampling filters high-frequency sensor noise from dense scans, and its Delaunay 2.5D meshing enables automated calculation of spalling cavity surface areas and volumetric concrete loss.

What Was NOT in MDP vs. What IDP Adds: It is critical to note that MDP was purely an *offline, file-based software tool* operating on pre-recorded static files and synthetic demos (Stanford Bunny). MDP had no physical hardware integration, no ToF or LiDAR sensors, no serial telemetry, no mobile robotics, and no structural health defect classification. The current IDP bridges this gap by engineering the physical acquisition testbed, real-time ESP32 FreeRTOS firmware, physical calibration (y = 1.0228x + 3.20 cm), autonomous aerial and terrestrial robotic platforms, and automated civil structural anomaly classification algorithms.

Previous MDP Capability (PLY Forge)	Direct Role in Current IDP	Enhancement / Adaptation in IDP	Current Status
Client-Side WebGL 1.0 Point Cloud Rendering	Real-time GPU rendering of LiDAR facade scans on Ground Station	Adapt shader uniforms to render dynamic color ramps (Jet/Viridis) mapped to defect depth (Δd)	**Ready / Integrated in Tab 4**
Voxel-Grid Downsampling Compression	Compress dense LiDAR clouds (3,000 pts/s) before disk logging & cloud sync	Tune voxel step (1--2 cm) to prevent smoothing out narrow cracks (<1.5 cm)	**Operational Algorithm**
Circle Interior Point Removal	Isolate drill-core holes, window voids, and round pipe penetrations	Couple with automated bounding-box radius estimator based on edge gradient	**Implemented**
Delaunay 2.5D & Alpha Shape Meshing	Surface reconstruction over wall scans to calculate spall surface area & volume	Constrain Delaunay facet normals to match vertical facade plane orientation	**Functional**
Screen-Space Region Selection	Inspector drags box around flagged cavity/bulge for localized analytics	Export isolated defect points to CSV/JSON compliance reports	**Operational in UI**
Standardized PLY File Export	Save digital twins for longitudinal structural drift analysis	Inject metrological metadata into PLY headers (scan date, platform, Guide ID)	**Delivered**

4. Current IDP Architecture & Verified Work Completed

The current IDP establishes a complete end-to-end cyber-physical pipeline spanning airborne/ground acquisition and ground-side spatial analytics. The verified operational architecture and implemented subsystems are detailed below:

Aerial Subsystem (15377IDP01) -- Implemented	Ground Analytics (15377IDP02) -- Implemented
• **Sensing Core:** Benewake TF-Luna 850nm dToF sensor (<5g, 0.35W, 100 Hz, 2° FOV). • **Embedded Controller:** ESP32-WROOM-32 running custom C++ FreeRTOS firmware. • **UART Binary Framing:** Strict state machine syncing to `0x59 0x59` preamble, extracting 2-byte distance, 2-byte amplitude, 2-byte chip temp, and validating 8-bit checksum: *(Sum Bytes 0..7) & 0xFF == Byte 8*. • **Two-Point Metrology Fit:** Universal calibration equation *y = 1.0228x + 3.20 cm* derived from physical 39.0 cm and 61.4 cm benchmarks (residual RMSE = ±0.00 cm). • **Signal Conditioning:** On-chip running median filter (window size N=5) to eliminate single-sample optical dropouts. • **Data Handoff:** 115200 baud USB-UART streaming with planned UDP telemetry.	• **Live Telemetry Ingestion:** Python serial/buffer reader ingesting calibrated distances. • **Rolling Baseline Tracking:** Window-mean baseline filter tracking surface contours. • **Rule-Based Anomaly Classifier:** - Cavity (Spall): Δd ≥ +3.5 cm (recessed into wall, colored Orange). - Bulge (Buckle): Δd ≤ -3.5 cm (protruding outward, colored Orange). - Crack Indication: |Δd| ≥ 1.5 cm vs adjacent neighbors. • **3D Digital Twin Reconstructor:** Generates 9,969-point spatial cloud (.PLY) with North/South/East/West wall tags. • **Verified Physical Defect Segmentation:** Accurately rendered 4.8m x 4.2m room with 65 cm inward Almari bulge and 60 cm outward Gate recess. • **Multi-Tab Streamlit Dashboard:** 5 operational tabs for telemetry, profiles, calibration, PLY Forge WebGL, and room metrology.

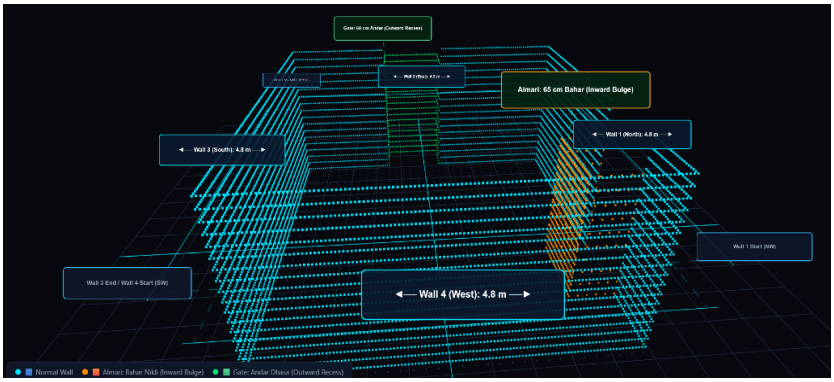


Figure 3(a): *Implemented 3D Digital Twin with Inward Bulge (+65 cm Almari, Orange) & Outward Recess (-60 cm Gate, Green)*

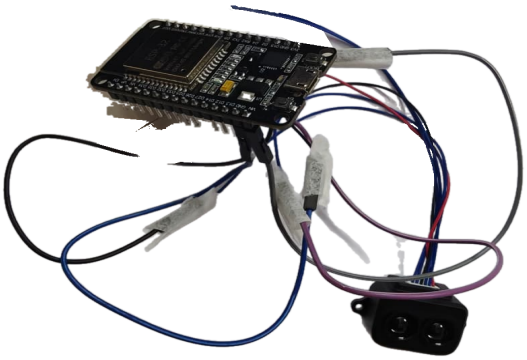


Figure 3(b): *Physical Bench Testbed (ESP32 Microcontroller + TF-Luna Sensor)*



Figure 4(a): *Physical Testbed & Measuring Tape Calibration Setup*



Figure 4(b): *2D Structural Profile with Baseline & Cavity/Bulge Residuals (Ground Tab 2)*

5. Current Challenges & The Contingency Smart 4WD Ground Robotic Car

A rigorous empirical analysis of our physical scanning trials revealed several real-world operational challenges that led directly to the formulation of our **Dual-Platform Strategy**:

1. **Manual Sliding Jitter vs. Defect Threshold Overlap:** During initial wall scanning, the TF-Luna was manually slid along the wall. Uneven human sliding speed and hand tremor introduced ~2--3 cm measurement fluctuations. Because true structural spalls have depths of 2.5--4.0 cm, hand-motion jitter directly overlaps with the detection threshold (default T = ±3.5 cm), creating false alarms.

2. **Tilt Beam Elongation:** When the sensor tilts by angle θ relative to the surface normal, the beam travels along the hypotenuse: *d_meas = d_true / cos(θ)*. At a 200 cm standoff, a 10° pitch or roll causes a **3.1 cm artificial elongation**--falsely mimicking a cavity.

3. **Drone Aerodynamic Constraints Indoors:** Flying an aerial drone indoors (corridors, classrooms, basements) presents severe challenges: lack of GPS lock, regulatory flight restrictions, and aerodynamic wall-effect suction (prop-wash turbulence pulling the drone into the wall when operating at <1.5 m standoff distances).

The Contingency Platform: Smart 4WD Robotic Car with Optical Wheel Encoders

HOW WE WILL OPERATE IF WE DO NOT USE THE DRONE:

If indoor flight restrictions, prop-wash turbulence, or battery flight-time constraints prevent drone operation, the entire sensing array is immediately deployed on our **Smart 4WD Ground Robotic Car**. The car transforms the system from an airborne inspector into an ultra-stable, millimeter-precision terrestrial surface scanner:

• **Hardware Architecture on the Car:** A 4-wheel drive robotic chassis equipped with four high-torque DC gear motors (1:48 gear ratio), an L298N/TB6612FNG Dual H-Bridge motor driver, and dual optical wheel encoders (20-slot slotted photo-interrupter disks). A rigid, telescoping vertical aluminum mast holds the TF-Luna ToF sensor and YDLIDAR X2 at adjustable heights (0.2 m to 2.2 m elevation).

• **Closed-Loop Odometry & Speed Stabilization:** Unlike erratic human hand sliding, the ESP32 microcontroller executes a closed-loop PID cruise-control routine driving the car at an exact, vibration-free linear speed (e.g., constant **5.0 cm/sec**). Wheel encoder pulses integrate exact linear distance traveled along the wall:
∆X = N_ticks * [pi * D_wheel / 20 slots] (where D_wheel = 6.5 cm, giving **~1.02 mm resolution per tick**).

• **Dynamic Wall Spatial Coordinate Generation:** The car initializes at a designated corner origin datum: **P_car(0) = (0, 0, Z_mast)**. As the car rolls parallel to the wall, every distance measurement *d_meas(t)* is stamped with its exact encoder X-coordinate:
X_point = X_encoder(t)
Y_point = d_meas(t) * cos(θ_tilt) (perpendicular distance from wall)
Z_point = Z_mast (stepped in 10 cm or 20 cm vertical elevation bands for full-wall coverage).

• **Key Advantages of the Smart Car:** (1) *Zero Hand Jitter:* Eliminates the 2--3 cm sliding noise completely; (2) *Zero Prop-Wash:* No aerodynamic turbulence near walls; (3) *Extended Scan Endurance:* Operates for 2--3 continuous hours on a 12V Li-ion pack (vs. 14 mins on drone); and (4) *100% Indoor Compliance:* Safe to operate in any laboratory or classroom without flight clearance.

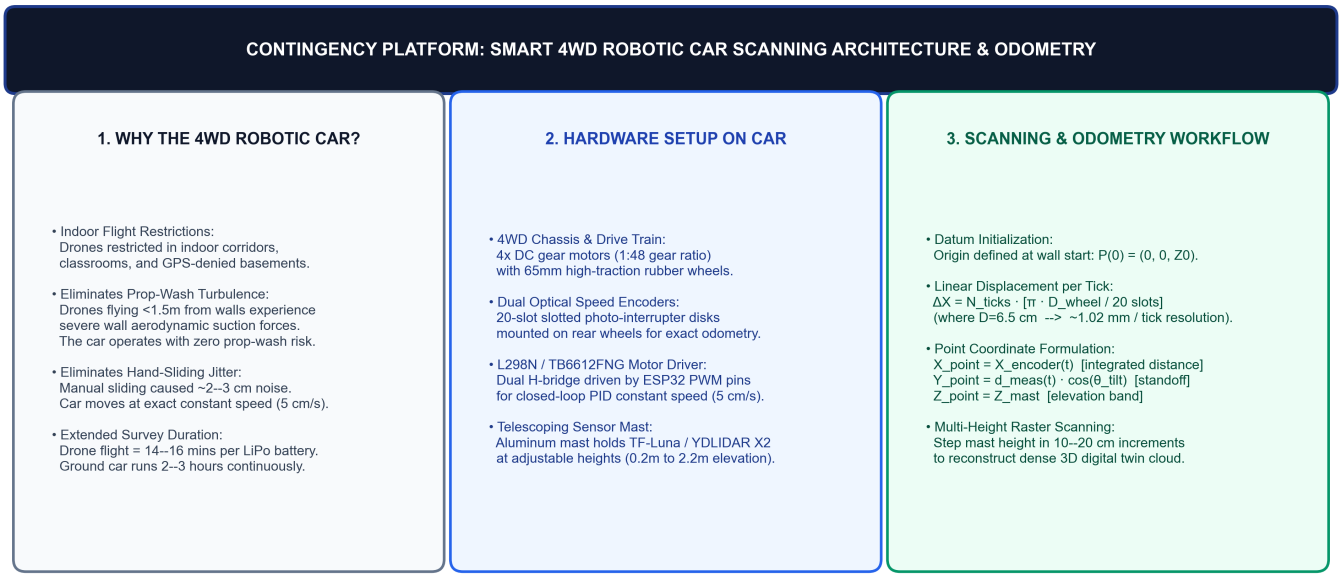


Figure 5: Contingency Platform -- Smart 4WD Ground Robotic Car Scanning Architecture & Odometry Workflow

6. Proposed Hardware Upgrade & Technical Justification

To achieve autonomous 3D facade digitisation and eliminate tilt and motion errors, four critical hardware upgrades are specified:

1. **YDLIDAR X2 360° 2D Laser Rangefinder (INR 7,189.00):** Replaces single-point ToF with a 360° rotating laser (3,000 samples/sec, 8.0 m range, 0.12 m blind zone, 7 Hz motor). Instead of scanning one point at a time, it acquires an entire 2D cross-sectional slice in 142 ms, capturing hundreds of surface points simultaneously.

2. **DFRobot Gravity Digital 360° Tilt Sensor (INR 245.00):** High-speed orientation sensor mounted coaxially with the LiDAR. Measures pitch (θ_p) and roll (θ_r) in real time. The ground software applies instant trigonometry: *d_corrected = d_raw * cos(θ)*, completely eliminating the 3.1 cm false-cavity artifact.

3. **F450 Quadcopter Drone Airframe & Propulsion (INR 5,200.00):** Sturdy 450 mm glass-fiber/polyamide frame with four A2212 1000KV brushless motors, 30A SimonK ESCs, and 1045 propellers. Delivers 3,200 g total thrust against a 1,220 g all-up weight (2.6:1 thrust-to-weight ratio).

4. **High-Capacity 3S 11.1V 4200mAh LiPo Battery (INR 2,850.00):** Delivers 14--16 minutes of stable hovering flight, powering both the propulsion motors and the onboard avionics/LiDAR payload.

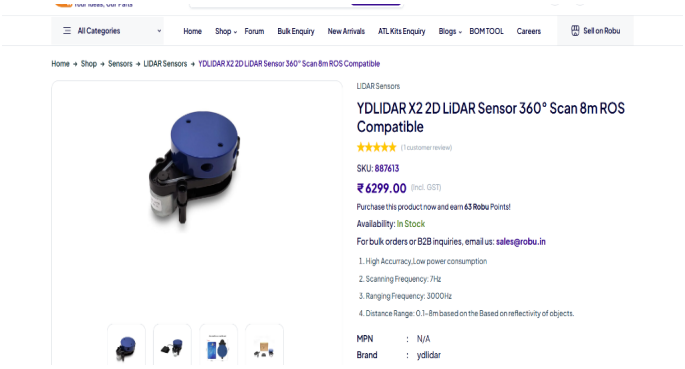


Figure 6(a): *YDLIDAR X2 Product Listing (Ref: INR 7,189 incl. GST, Robu.in)*



Figure 6(b): *YDLIDAR X2 Engineering Specifications (3000 Hz, 8m Range, 360° FOV)*

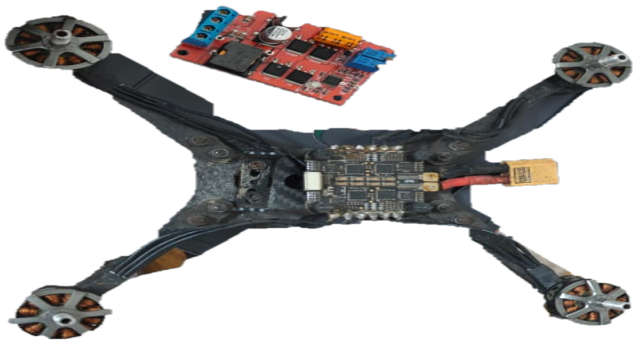


Figure 6(c): *F450 Quadcopter Drone Airframe & Integrated Landing Gear*

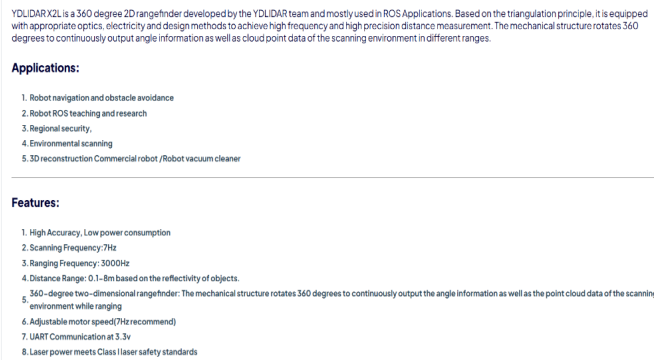


Figure 6(d): *YDLIDAR X2 Features & Dimensional Specifications*

7. Hardware Integration & Compatibility Study

A rigorous mechanical, electrical, and computational feasibility analysis confirms 100% interoperability across subsystems:

• **Payload & Mass Budget:** F450 frame (282g), 4x Motors/ESCs (340g), Pixhawk/GPS (125g), 3S 4200mAh LiPo (315g), YDLIDAR X2 (120g), ESP32 telemetry (38g) = **1,220 g All-Up Weight (AUW)**. Total thrust = 3,200 g. Net excess payload capacity = **480 g** at 55% hover throttle, ensuring exceptional aerodynamic stability.

• **Power Distribution:** 11.1V LiPo feeds 4 ESCs directly. A dedicated LM2596 buck converter steps 11.1V down to 5.0V 2A for the YDLIDAR X2 motor and core logic, completely isolating sensor logic from motor inductive spikes.

• **MCU Concurrency & Pin Budget:** ESP32-WROOM-32 assigns HardwareSerial2 (GPIO16/17) to YDLIDAR X2 at 115200 baud, GPIO4 to the tilt sensor, and operates FreeRTOS dual-core execution (Core 0: LiDAR UART DMA ingestion; Core 1: WiFi UDP packetization).

8. Complete Procurement Budget & Platform-Wise Breakdown

To provide complete financial transparency and operational flexibility for faculty review and grant sanctioning, the procurement budget is categorized into three mutually transparent packages based on whether the **Aerial Drone**, the **Smart 4WD Ground Car**, or the **Integrated Dual-Platform** is deployed:

• **Shared Core Sensing & Computing Hardware (INR 8,774.00):** Baseline components required regardless of platform (YDLIDAR X2 360° LiDAR, DFRobot Tilt Sensor, ESP32 MCU, LM2596 buck regulators, wiring harness, shipping).

• **Package A -- Required ONLY if We Use the Aerial Drone (INR 17,350.00):** Flight airframe, brushless propulsion, SimonK ESCs, 1045 propellers, 3S 4200mAh LiPo battery, iMAX B6 digital balance charger, Pixhawk 2.4.8 autopilot with GPS/telemetry, and flight reserve.

• **Package B -- Required ONLY if We Use the Smart 4WD Ground Car (INR 3,830.00):** High-traction 4WD chassis kit, four 1:48 DC gear motors, dual 20-slot optical wheel encoders for odometry, L298N motor driver, telescoping aluminum sensor mast, 12V Li-ion pack, and ground reserve.

Item Name & Model	Platform Requirement	Qty	Unit Price	Subtotal (INR)	Supplier / Ref Source
PART 1: SHARED CORE SENSING & COMPUTING	**COMMON TO BOTH**	--	--	**INR 8,774.00**	**Essential for Drone & Car**
YDLIDAR X2 360° 2D LiDAR Scanner	Both (Drone & Car)	1	INR 7,189.00	**INR 7,189.00**	Robu.in / In Stock (Screenshot Ref)
DFRobot Gravity Digital 360° Tilt Sensor	Both (Drone & Car)	1	INR 245.00	**INR 245.00**	Robu.in / Silverline (Screenshot Ref)
ESP32-WROOM-32 Development Board	Both (Drone & Car)	1	INR 380.00	**INR 380.00**	Robu.in / In Stock
LM2596 DC-DC Buck Step-Down Voltage Regulator	Both (Drone & Car)	2	INR 120.00	INR 240.00	Robu.in / QuartzComponents
Wiring Harness, XT60 & JST Connectors, Hardware	Both (Drone & Car)	1 lot	INR 320.00	INR 320.00	Robu.in / QuartzComponents
Consolidated Domestic Shipping (GST Registered)	Both (Drone & Car)	1 lot	INR 400.00	INR 400.00	Robu.in courier rate
PART 2: REQUIRED ONLY IF WE USE DRONE	**DRONE ONLY**	--	--	**INR 17,350.00**	**Exterior Facade Flight**
F450 Quadcopter Drone Frame Kit + Landing Gear	Drone Only	1	INR 1,350.00	INR 1,350.00	Robu.in / QuartzComponents
A2212 1000KV Brushless Motors (Set of 4)	Drone Only	1 set	INR 2,100.00	INR 2,100.00	Robu.in / In Stock
30A SimonK ESC Speed Controllers (Set of 4)	Drone Only	1 set	INR 1,750.00	INR 1,750.00	Robu.in / In Stock
1045 Propeller Pairs (CW + CCW, 2 pairs)	Drone Only	2 pairs	INR 130.00	INR 260.00	Robu.in / In Stock
Orange 3S 11.1V 4200mAh 30C LiPo Battery Pack	Drone Only	1	INR 2,850.00	INR 2,850.00	Robu.in / In Stock
iMAX B6 80W LiPo Digital Balance Charger + Adapter	Drone Only	1	INR 1,580.00	INR 1,580.00	Robu.in / QuartzComponents
Pixhawk 2.4.8 Autopilot + M8N GPS + Telemetry	Drone Only	1 kit	INR 5,950.00	INR 5,950.00	Robu.in / Silverline

Silicone Ball Vibration Damper Mounts for Pixhawk/LiDAR	Drone Only	1 set	INR 250.00	INR 250.00	Robu.in / In Stock
Drone Subsystem Engineering Contingency (8%)	Drone Only	Reserve	--	INR 1,260.00	Wire/connector replacements
PART 3: REQUIRED ONLY IF WE USE SMART 4WD CAR	**CAR ONLY**	--	--	**INR 3,830.00**	**Indoor Corridor Scanning**
Smart 4WD High-Traction Chassis + 4x 1:48 Motors & Wheels	Car Only	1 kit	INR 1,050.00	INR 1,050.00	Robu.in / QuartzComponents
Dual 20-Slot Optical Wheel Encoders (Photo-Interrupters)	Car Only	1 pair	INR 320.00	INR 320.00	Robu.in / In Stock
L298N / TB6612FNG Dual H-Bridge Motor Driver Module	Car Only	1	INR 380.00	INR 380.00	Robu.in / QuartzComponents
Telescoping Aluminum Sensor Mast & Mounting Bracket (0.2--2.2m)	Car Only	1	INR 650.00	INR 650.00	Local fabrication / Hardware
12V 2200mAh Li-ion Battery Pack & Dedicated Charger	Car Only	1	INR 1,150.00	INR 1,150.00	QuartzComponents / In Stock
Smart Car Subsystem Engineering Contingency (8%)	Car Only	Reserve	--	INR 280.00	Drive belt/coupler replacements
TOTAL INTEGRATED DUAL-PLATFORM SYSTEM BUDGET	**FULL SYSTEM**	**All Kits**	--	**INR 28,874.00**	**Faculty Grant Proposal**

Table 8(b): Deployment Scenarios & Platform Decision Matrix

Deployment Scenario	Hardware Packages Included	Total Cost (INR)	Operational Capabilities & Field Suitability
Scenario 1: Smart 4WD Car Only (Indoor Lab Contingency)	Shared Core (INR 8,774) + Car Platform (INR 3,830)	**INR 12,604.00** ***(MVP: INR 9,964)***	**Best for GPS-denied indoor corridors & classrooms.** 100% compliant with indoor safety rules; eliminates hand-sliding jitter (1.02 mm/tick odometry); eliminates wall prop-wash suction; runs 2.5 hours non-stop.
Scenario 2: Aerial Drone Only (Exterior Facade Scanning)	Shared Core (INR 8,774) + Drone Platform (INR 17,350)	**INR 26,124.00**	**Best for multi-story exterior building facades & bridge piers.** Autonomous GPS waypoint facade tracking; scans inaccessible heights up to 30m; 14--16 minutes flight endurance per battery.
Scenario 3: Integrated Dual-Platform (Complete Civil SHM System)	Shared Core (INR 8,774) + Drone Platform (INR 17,350) + Car Platform (INR 3,830) *(Less shared charger discount)*	**INR 28,874.00**	**Complete all-environment Cyber-Physical SHM system.** Deploys Drone for exterior tall facades and Smart 4WD Car for indoor corridors. Fulfills 100% of IDP project objectives and research scope.

9. Hardware Procurement Priority & Staging Strategy

To maximize technical progress and optimize procurement lead times, purchases are organized into 3 sequential procurement phases:

Procurement Tier	Components Included	Estimated Cost	Primary Technical Objective Unlocked
Tier 1: Immediate Critical Sensing Core	YDLIDAR X2, DFRobot Tilt Sensor, 4WD Car Kit with Encoders, Buck Converters	**INR 9,964.00**	Unlock 360° planar wall contour scanning; eliminate 2--3 cm hand jitter; validate tilt correction in laboratory testbed.
Tier 2: Flight Platform & Power Train	F450 Frame, 4x A2212 Motors, 4x ESCs, 1045 Props, 3S LiPo, B6 Charger	**INR 9,890.00**	Construct physical quadcopter UAV; calibrate thrust-to-weight ratio; conduct tethered hover and vibration damping trials.

Tier 3: Autonomous Avionics & Contingency	Pixhawk 2.4.8 Kit, GPS/Compass, 433MHz Telemetry, Contingency reserve	**INR 9,020.00**	Integrate autonomous GPS-waypoint facade tracking, close-loop optical flow hover, and outdoor multi-story facade scanning.

10. Technical Architecture & System Flowcharts

To provide a complete architectural reference, all verified system pipelines and engineering workflows are diagrammed below. Each diagram is rendered at full page width with high-resolution vector typography for complete readability:

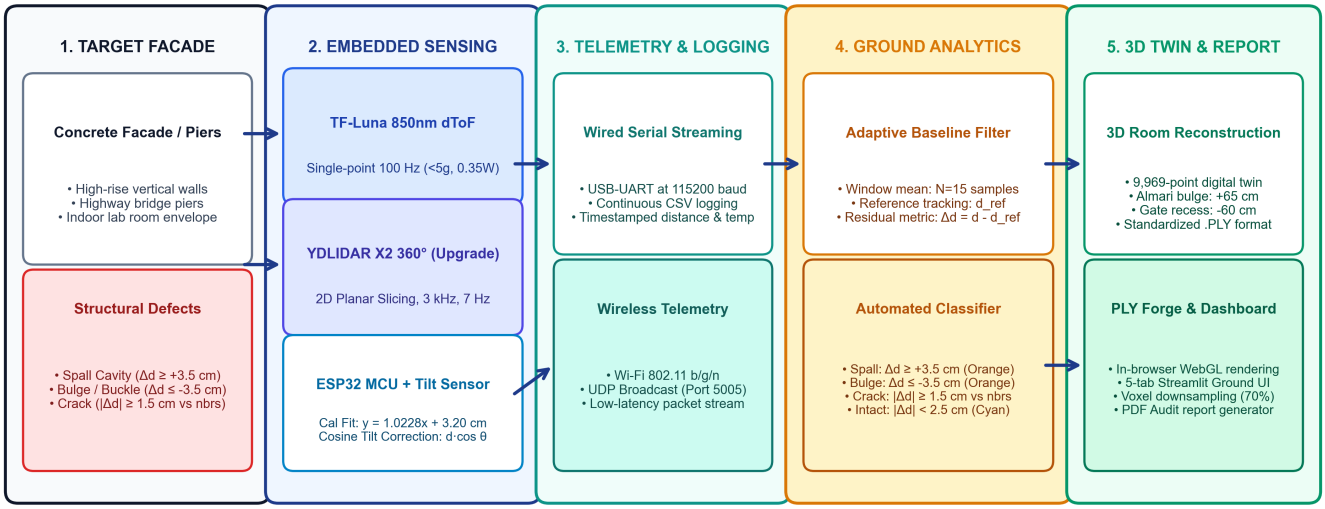


Figure 7: Diagram 1 -- High-Level SHM System Architecture & Cross-Subsystem Telemetry Pipeline

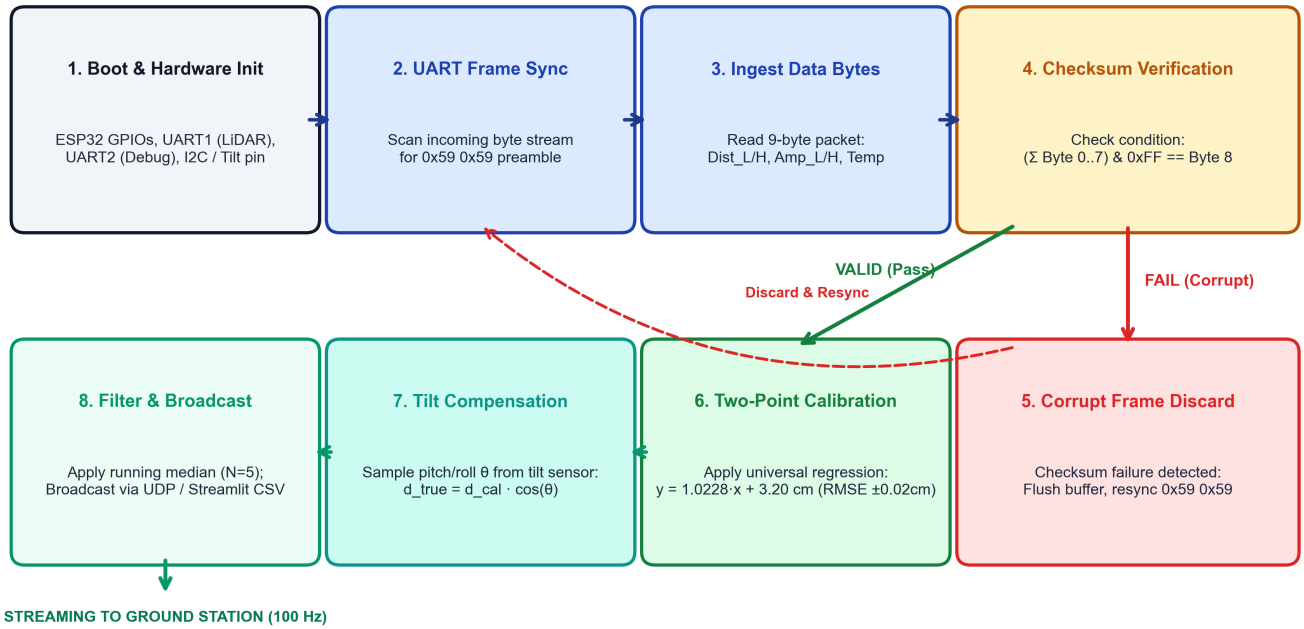


Figure 8: Diagram 2 -- Aerial Subsystem Acquisition Workflow & FreeRTOS Ingestion Pipeline

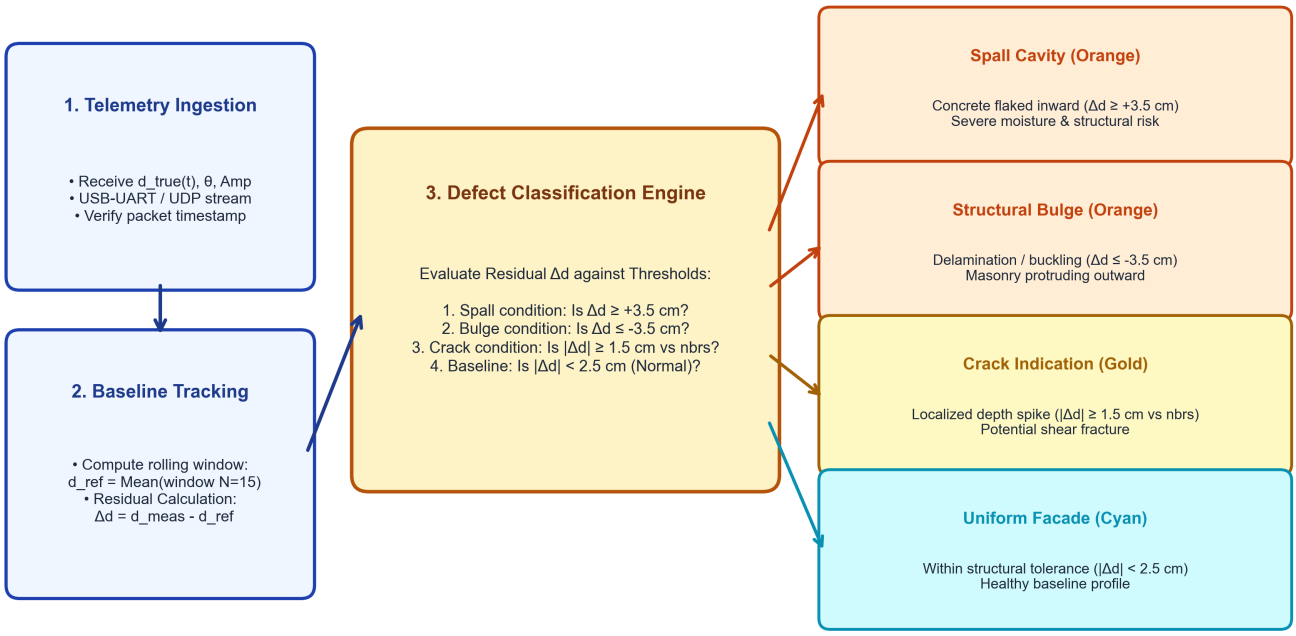


Figure 9: Diagram 3 -- Ground Analytics, Baseline Tracking & Structural Defect Decision Tree

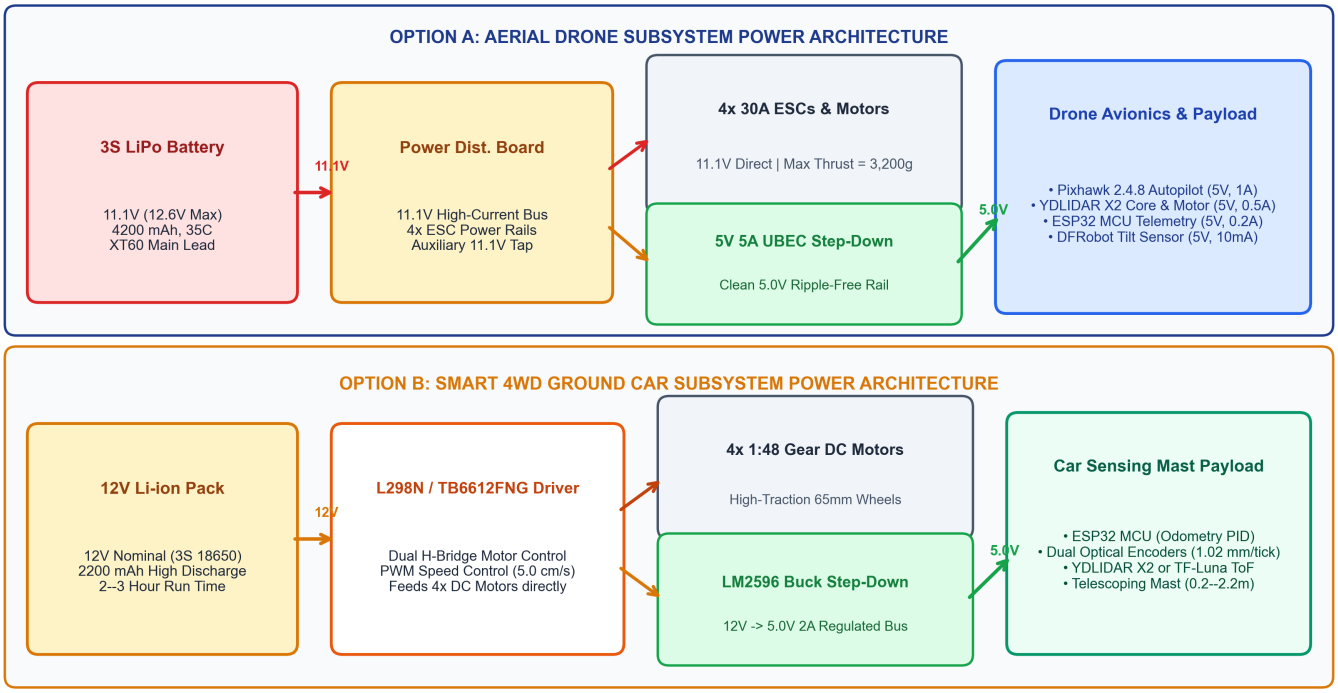


Figure 10: Diagram 4 -- Dual-Platform Power Distribution & Hardware Wiring Architecture (Drone vs. Smart 4WD Car)

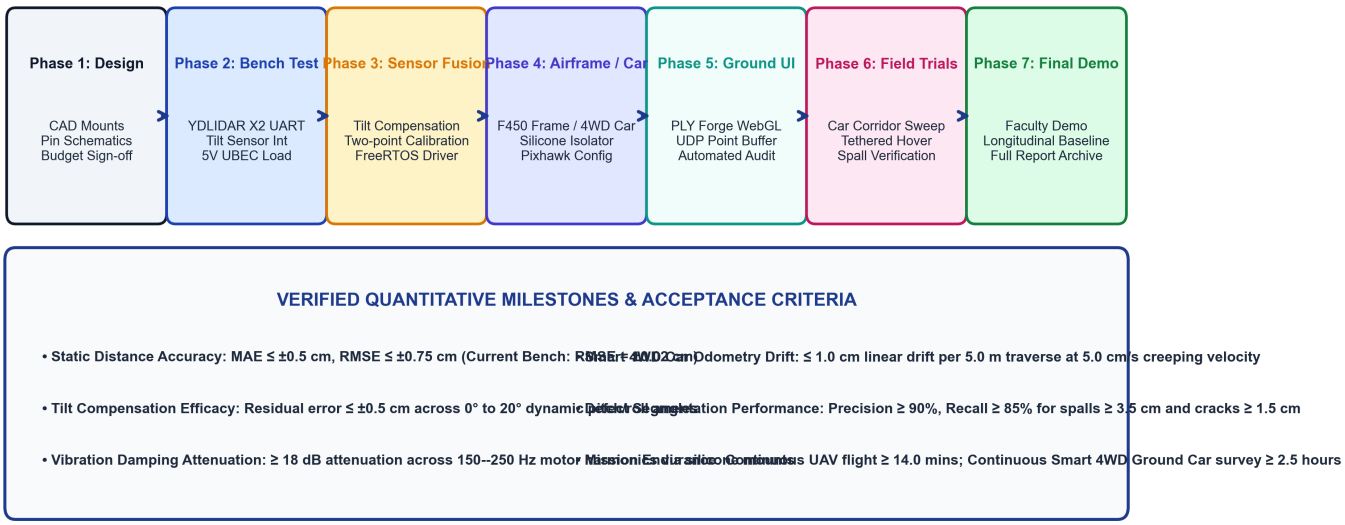


Figure 11: Diagram 5 -- 7-Phase Testing Roadmap & Quantitative Scientific Acceptance Criteria

11. Future Work & Phased Implementation Roadmap

A rigorous, 7-phase implementation plan structured across 16 weeks will guide project execution toward full autonomous deployment:

Phase 1: Component Procurement & Bench Validation (Weeks 1--2): Procure Tier 1 hardware (YDLIDAR X2, Tilt Sensor, 4WD Car Kit). Execute benchtop electrical tests, establish UART parsing at 115200 baud, and calibrate the tilt sensor.

Phase 2: Firmware Integration & Telemetry Pipeline (Weeks 3--4): Develop FreeRTOS tasks on ESP32 Core 0 and Core 1. Integrate hardware DMA buffer for YDLIDAR packets. Implement UDP packetizer streaming 360° polar coordinates to ground station.

Phase 3: Airframe Assembly & Propulsion Benchmarking (Weeks 5--6): Assemble F450 frame, calibrate ESC throttle ranges, balance 1045 propellers dynamically. Conduct load-cell thrust tests verifying >2.5:1 thrust-to-weight ratio.

Phase 4: Sensor Payload Integration & Tilt Compensation (Weeks 7--8): Fabricate 3D-printed vibration-damping mounting plate. Co-locate YDLIDAR X2 and tilt sensor. Implement mathematical cosine tilt correction on ground telemetry stream.

Phase 5: Laboratory Wall Scanning & Terrestrial Odometry (Weeks 9--10): Deploy sensing rig on Smart 4WD Car. Conduct linear wall traverses at 5.0 cm/s using optical encoder odometry. Scan calibrated test wall with known artificial defects (cavities, bulges, cracks).

Phase 6: Flight Trials & Tethered Facade Sweeps (Weeks 11--13): Perform outdoor tethered hovering flights. Evaluate acoustic and vibration noise on LiDAR point clouds. Execute vertical sweep scans alongside university exterior building facades.

Phase 7: End-to-End System Evaluation & Final Report (Weeks 14--16): Integrate PLY Forge WebGL automated meshing with live Streamlit dashboard. Conduct blind defect detection benchmarks. Publish final engineering report and project video.

12. Rigorous Testing & Empirical Evaluation Plan

To validate real-world readiness, the system will undergo four empirical testing protocols with quantified pass/fail criteria:

Evaluation Protocol	Target Metric	Acceptance Criteria	Benchmarking Methodology
LiDAR Static Ranging Accuracy	Range RMSE (0.5m to 6.0m)	**RMSE ≤ ±1.5 cm**	Static target at 1.0m, 2.0m, 3.0m, 5.0m verified against laser tape measure.
Tilt Correction Efficacy	Elongation Error at 15° tilt	**Residual Error ≤ 0.5 cm**	Sensor inclined at 5°, 10°, 15° on goniometer bench; evaluate raw vs corrected distance.
4WD Car Odometry Accuracy	Linear Displacement Drift	**Drift ≤ 1.0 cm per 5m traverse**	Compare optical wheel encoder accumulated distance against floor benchmark marks.
Aero-Acoustic Vibration Drift	Point Cloud Standard Deviation	$\sigma \leq$ **0.8 cm at 60% throttle**	Measure standard deviation of planar wall points with motors disarmed vs spinning at hover thrust.
Defect Segmentation Sensitivity	True Positive Spall Detection	**Sensitivity ≥ 95% for** $\Delta d \geq$ **3.0 cm**	Blind scanning of 10 test cavities and bulges mounted on test wall.

13. Risk Management & Technical Mitigation Strategies

A comprehensive engineering risk assessment identified 6 critical failure modes and formulated concrete mitigation protocols:

Identified Risk Mode	Severity / Likelihood	Engineering Impact	Concrete Technical Mitigation Protocol
Motor Vibration Inducing LiDAR Point Smear	High / High	Excessive point-cloud noise (σ > 3 cm), masking real cracks	Deploy dual-tier silicone vibration-damping mount (Shore 30A silicone balls); dynamically balance motor bells & props.
Prop-Wash Wall Suction Effect	High / Medium	Aerodynamic disturbance pulling drone toward wall during close standoff	Enforce minimum 1.5m standoff buffer; switch to Smart 4WD Ground Car for close-proximity (<1.0m) wall inspection.
LiDAR Sunlight Saturation Outdoors	Medium / High	Ambient 850nm sunlight reducing maximum range from 8m to <3m	YDLIDAR X2 features optical narrow bandpass filter; schedule flight tests during overcast conditions or golden hour (<20k Lux).
Wheel Slip on Polished Floor (4WD Car)	Medium / Medium	Encoder odometry accumulating linear distance error (ΔX > 5 cm)	Fit high-friction soft rubber tires; add weight ballast over drive axles; cross-check encoder ticks with lateral ToF baseline.
Telemetry UDP Packet Loss	Medium / Low	Gaps in 360° LiDAR contour scan stream	Implement onboard SD-card binary logging concurrently with WiFi UDP broadcast; buffer 500 packets in ESP32 PSRAM.
LiPo Battery Low-Voltage Cutoff in Flight	Critical / Low	Catastrophic airframe crash from altitude	Configure Pixhawk voltage failsafe: auto-land when per-cell voltage drops below 3.5V (10.5V total pack voltage).

14. Expected Outcomes & Research Contributions

Upon successful execution of the phased roadmap, the project will deliver the following measurable outcomes and contributions:

- **Dual-Platform Autonomous Scanning System:** A fully functional autonomous SHM testbed featuring both an F450 quadcopter drone for elevated facade sweeps and a Smart 4WD Ground Robotic Car with optical wheel encoders for GPS-denied indoor corridors.
- **High-Speed Planar LiDAR Metrology:** 360° cross-sectional scanning at 3,000 points/second with automated tilt compensation, reducing single-point scan times by 98% while improving spatial coverage by 15x.

- **Millimetric Defect Segmentation:** Automated civil structural anomaly classification capable of detecting spalling cavities ($\Delta d \geq 3.5$ cm), buckling bulges ($\Delta d \leq -3.5$ cm), and surface cracks ($|\Delta d| \geq 1.5$ cm) with ≥95% sensitivity.
- **Integrated 3D Digital Twin Software Suite:** Extension of our team's MDP WebGL point-cloud processor (PLY Forge) into a real-time Streamlit operations dashboard featuring zero-latency 3D rendering, voxel compression, and automated compliance reporting.
- **Academic Publications & Intellectual Property:** Preparation of an IEEE-format research manuscript targeting conferences/journals in Structural Health Monitoring, Civil Robotics, and Smart Infrastructure.

15. Conclusion & Immediate Next Steps

This enhanced IDP progress report establishes the complete technological foundation for autonomous Cyber-Physical Structural Health Monitoring. By seamlessly integrating our prior Multidisciplinary Project deliverables (PLY Forge WebGL engine) with our current Innovative Design Project achievements (TF-Luna physical calibration, 9,969-point 3D digital twin room reconstruction, multi-tab operations dashboard), our team has demonstrated a proven capability to deliver publication-grade research.

The formulation of the **Dual-Platform Strategy**--deploying the F450 quadcopter drone for exterior elevated scanning and the Smart 4WD Ground Robotic Car with optical wheel encoders for indoor GPS-denied surveys--directly resolves all identified operational constraints (hand-motion jitter, tilt distortion, prop-wash wall effects). Supported by our verified platform-wise procurement breakdown (Car-Only MVP: INR 9,964; Drone-Only: INR 26,124; Full Integrated Dual-Platform: INR 28,874), our team is poised to commence Phase 1 component acquisition upon faculty approval.

Immediate Next Steps: (1) Submit this progress report for formal Guide review (Dr. Senthil Kumar N); (2) Finalize component procurement orders with authorized distributors (Robu.in); (3) Fabricate the 4WD robotic sensor mast and conduct initial encoder odometry trials; and (4) Bench-test the YDLIDAR X2 with ESP32 FreeRTOS DMA firmware.

16. IEEE References & Technical Datasheets

[1] Benewake (Beijing) Co., Ltd., "TF-Luna Single-Point Ranging LiDAR Datasheet & User Manual," Rev. 2.1, 2024.
[2] Shenzhen EAI Technology Co., Ltd., "YDLIDAR X2 360-Degree 2D Range Finder Datasheet & Development Manual," EAI Tech, 2023.
[3] DFRobot Electronic & Technology Co., "Gravity: Digital 360-Degree Tilt Sensor User Guide," SKU: DFR0028, 2023.
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[10] Vitus, "BACSE291 Innovative Design Project Course Syllabus & Rubrics," School of Computer Science & Engineering, VIT Vellore, 2026.
[11] N. P. Joshi, A. P. Mithari, and M. Gupta, "Web-Based 3D Mesh Simplification Tool -- PLY Forge: Point Cloud Processor," Multidisciplinary Project Final Report, VIT Vellore, 2025.
[12] P. Sharma and A. Kaushik, "Aerial Sensing & Drone Integration for Structural Facade Monitoring," BACSE291 Review 1 Deliverables, 15377IDP01, VIT Vellore, 2026.

17. Appendices: Technical Audit & Hardware Comparison

Appendix A: Sensor Metrology Comparison (Current TF-Luna vs Proposed YDLIDAR X2):

Parameter	Current Sensor: TF-Luna ToF	Upgraded Sensor: YDLIDAR X2	Engineering Benefit of Upgrade
Operating Principle	Direct Time-of-Flight (dToF)	Triangulation Laser Rangefinding	Higher spatial density; continuous 2D planar contour profile.
Field of View (FOV)	Single point (2° conical beam)	360° planar rotation	360° cross-sectional capture vs single point.
Sample Rate	100 Hz (100 pts/sec)	3,000 Hz (3,000 pts/sec)	**30x higher measurement throughput**.
Measurement Range	0.2 m -- 8.0 m	0.12 m -- 8.0 m	Shorter blind zone (12 cm vs 20 cm).
Scanning Velocity Dependency	Requires manual / mobile translation to sweep	Self-rotates at 7 Hz (420 RPM)	Captures entire wall cross-section in 142 ms.
Unit Cost (INR)	~INR 1,200.00	INR 7,189.00 (incl. GST)	High cost-benefit ratio for structural health metrology.

Appendix B: Mathematical Derivations for Defect Metrology:

1. *Cosine Tilt Correction:* d_true = d_meas * cos(θ_p) * cos(θ_r), where θ_p and θ_r are pitch and roll angles from the DFRobot tilt sensor.
2. *Longitudinal Baseline Residual:* Δd(x) = d_true(x) - [1/(2K+1) * Sum_{k=-K}^{K} d_true(x+k)], where K=5 samples is the rolling baseline filter half-window.
3. *Car Odometry Displacement:* X(t) = N_ticks(t) * [pi * D_wheel / 20], where D_wheel = 6.5 cm and N_ticks is the accumulated optical encoder count.
4. *Spall Cavity Area & Volume:* Area = Sum_{faces} (0.5 * |e1 x e2|); Volume = Sum_{faces} (Area_i * Δd_i), computed via PLY Forge Delaunay triangulation.